

VALVE POWER BOARD

RUBY - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

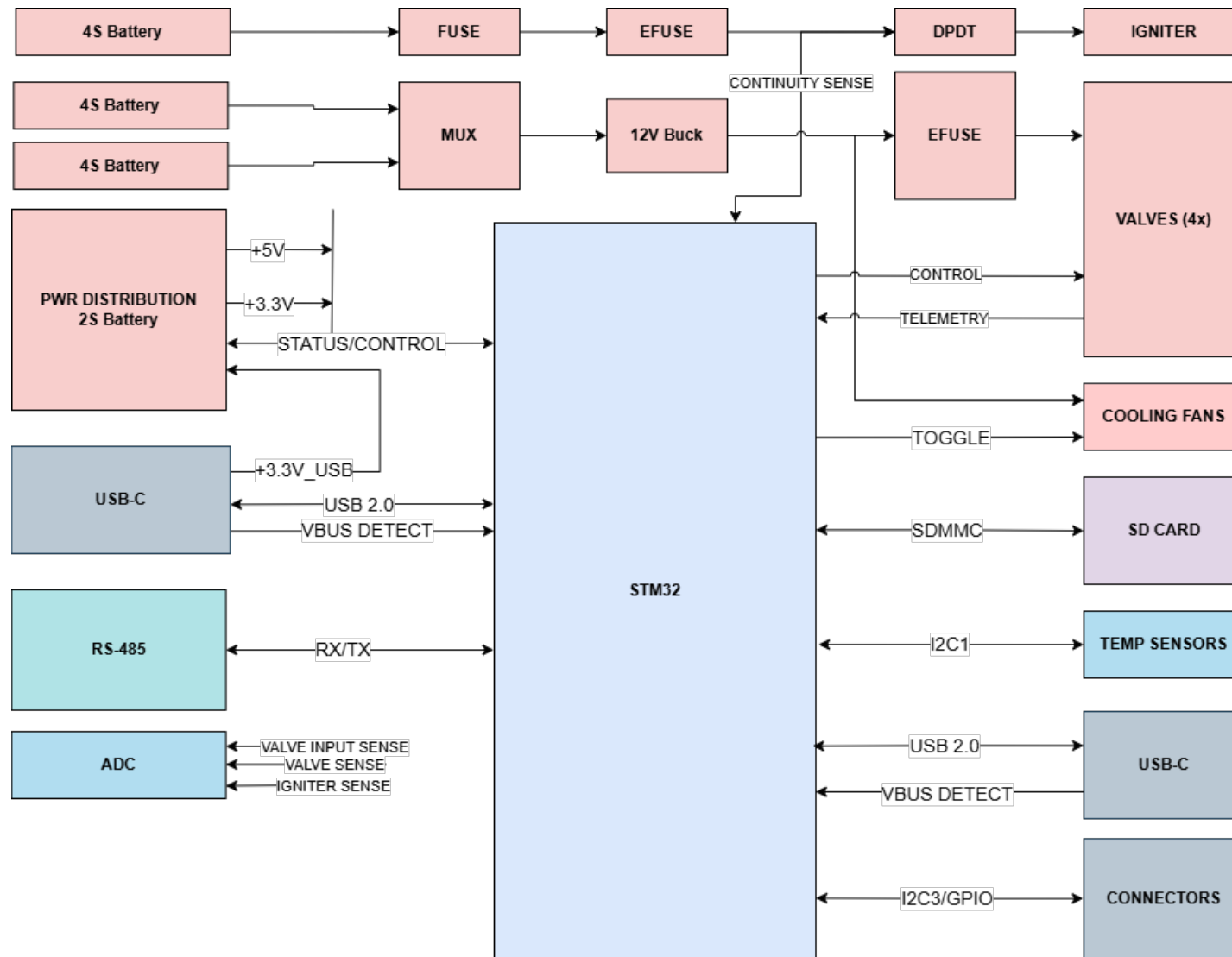
SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: top-level.kicad_sch

Title: TITLE PAGE

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

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Sheet: /Block Diagrams/
File: block_diagrams.kicad_sch

Title: BLOCK DIAGRAM

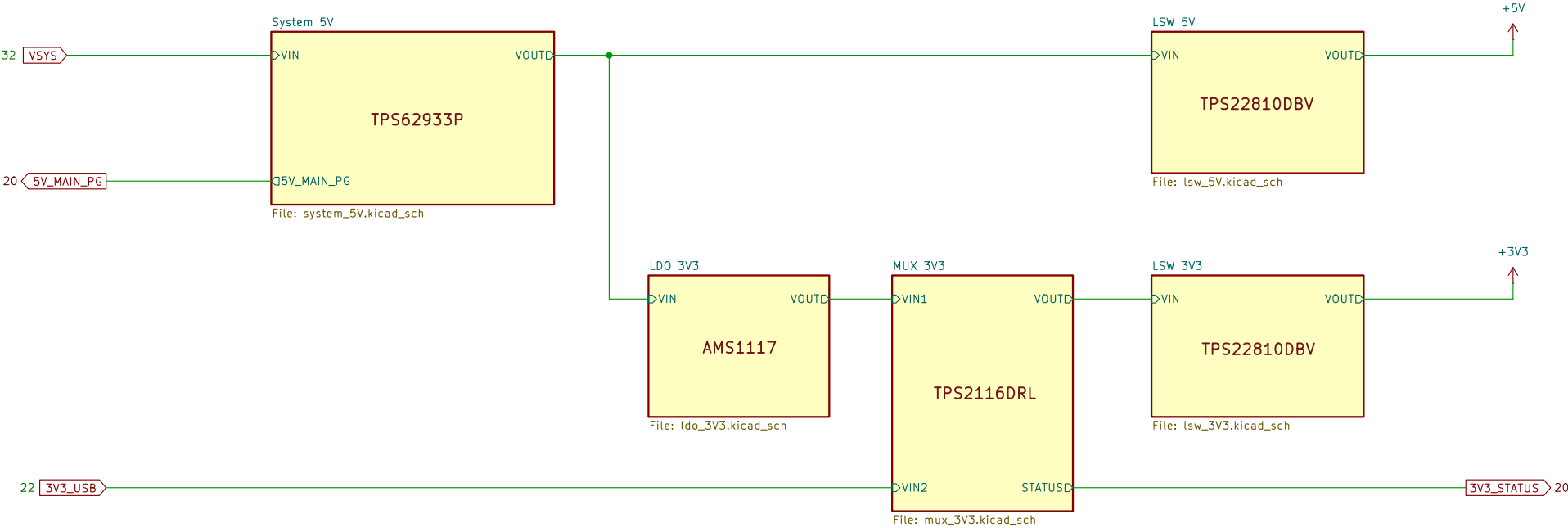
Size: A4 Date: 2026-10-08

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SYSTEM POWER



Sheet: /System Power/
File: system_power.kicad_sch

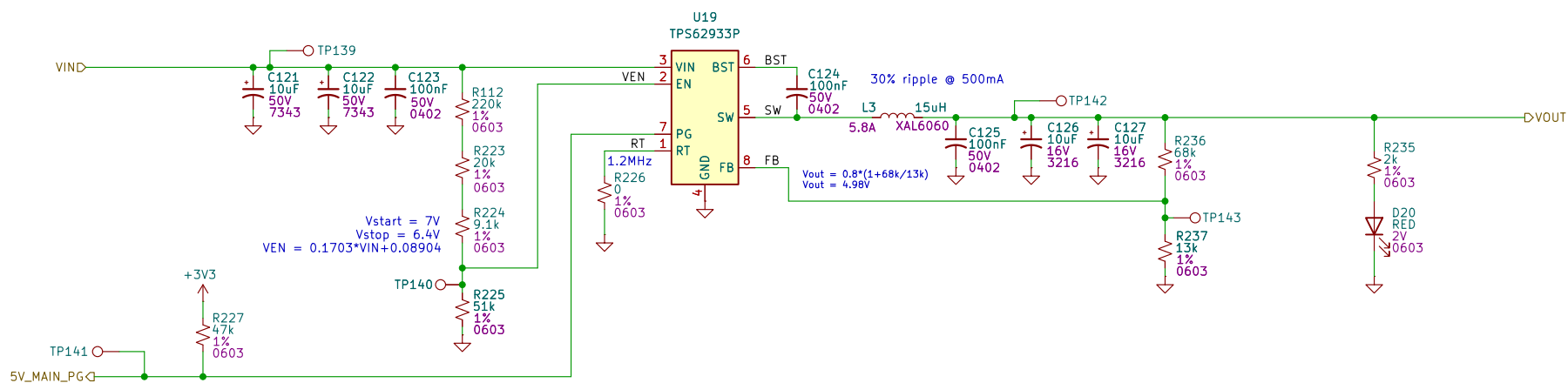
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Size: A4
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Sheet: /System Power/System 5V/
File: system_5V.kicad_sch

Title: SYSTEM 5V

Size: A4 Date: 2026-10-08

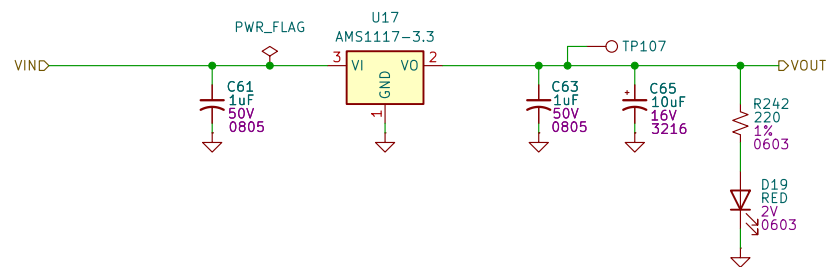
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Sheet: /System Power/LDO 3V3/
File: ldo_3V3.kicad_sch

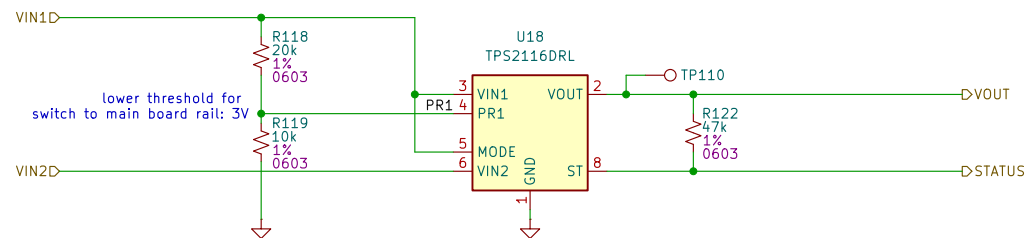
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Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

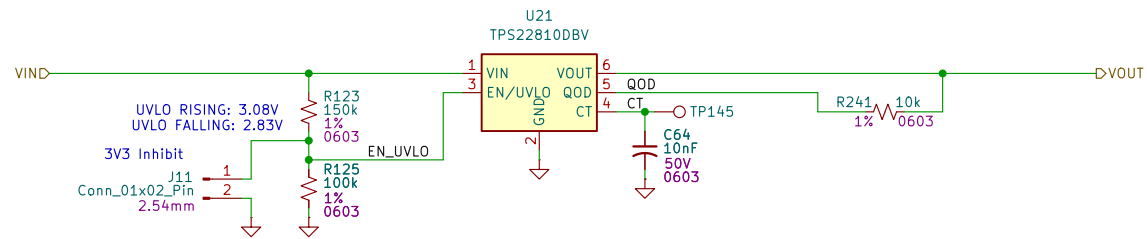
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Sheet: /System Power/MUX 3V3/
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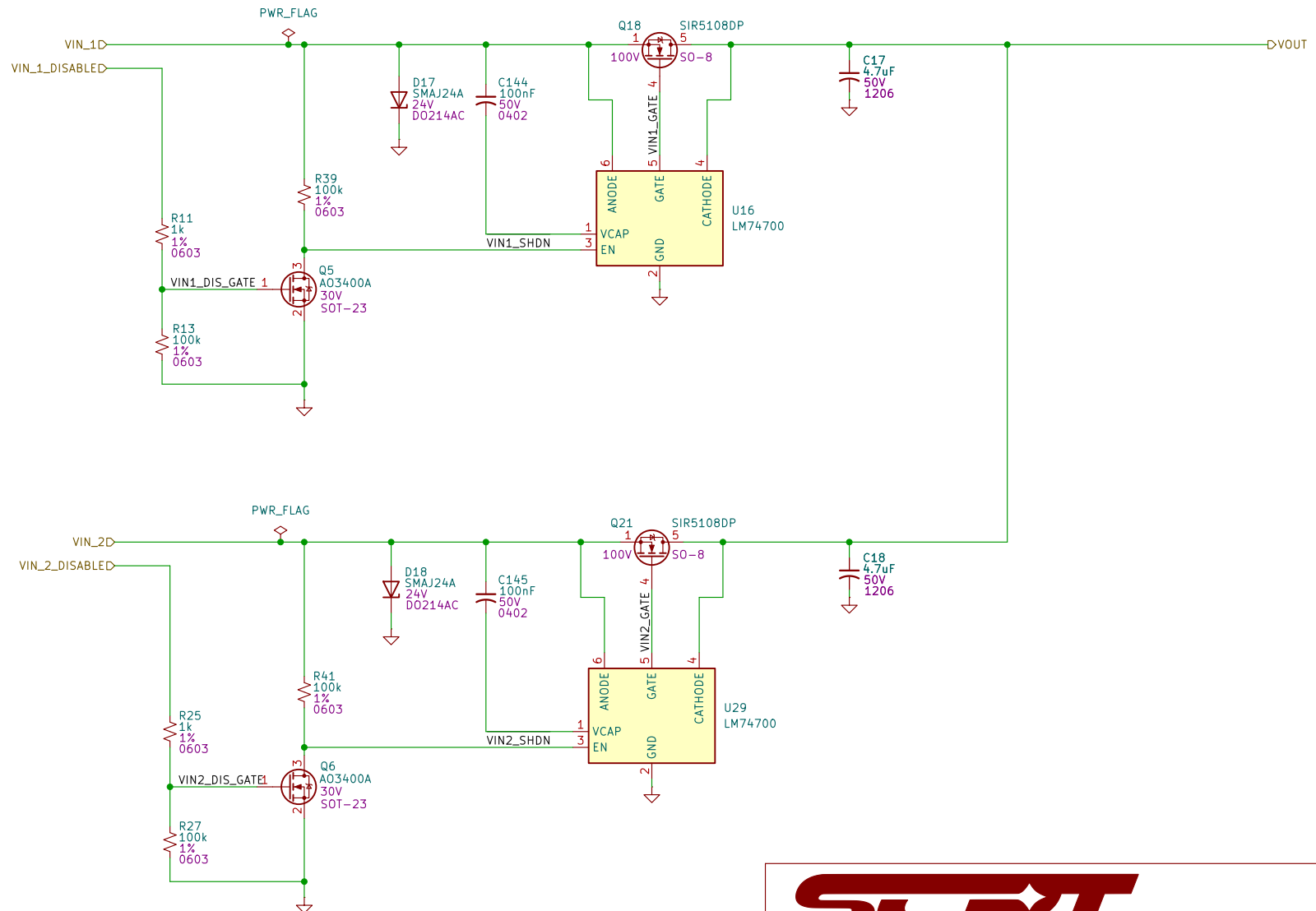
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Size: A4	Date: 2026-10-08	Rev: A
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Sheet: /System Power/LSW 3V3/ File: lsw_3V3.kicad_sch		
Title: LSW 3V3		
Size: A4	Date: 2026-10-08	Rev: A
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Protect against MOSFET avalanche



Sheet: /Valves/input_mux/
File: input_mux.kicad_sch

Title: INPUT MUX

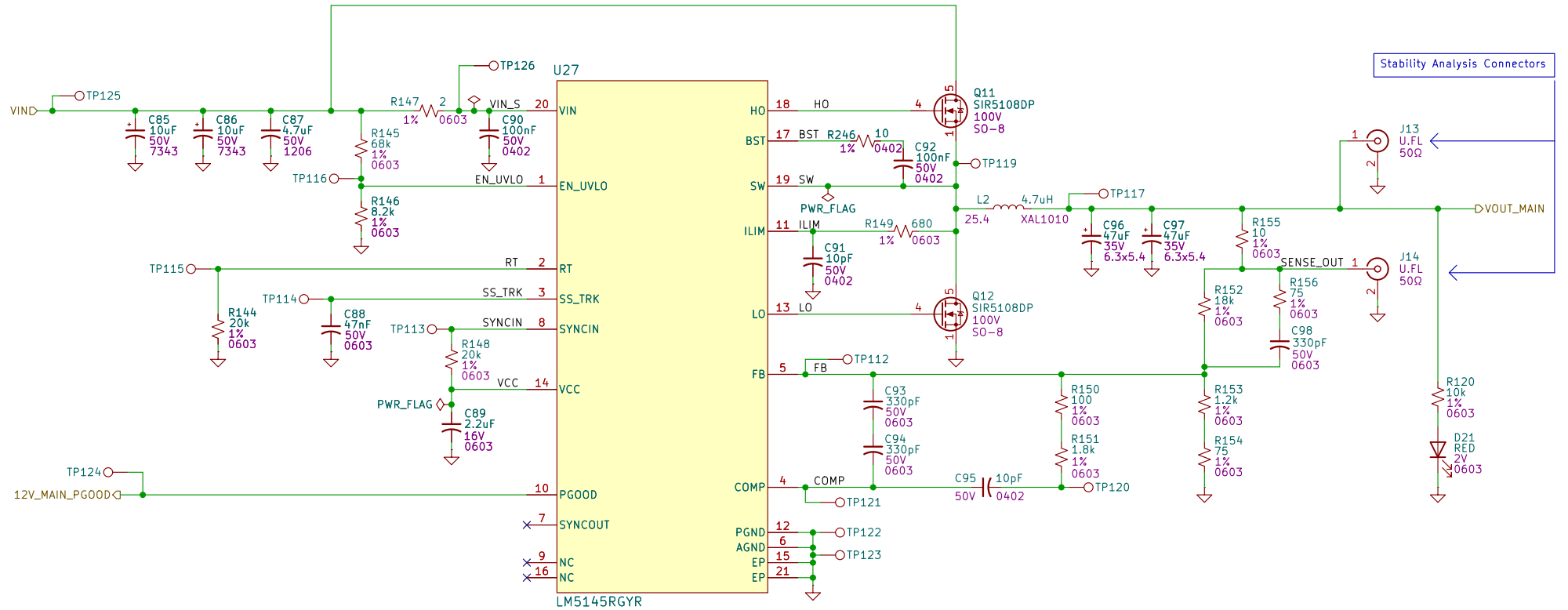
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12V MAIN



Sheet: /Valves/12V_main/
File: 12V_main.kicad_sch

Title: 12V MAIN

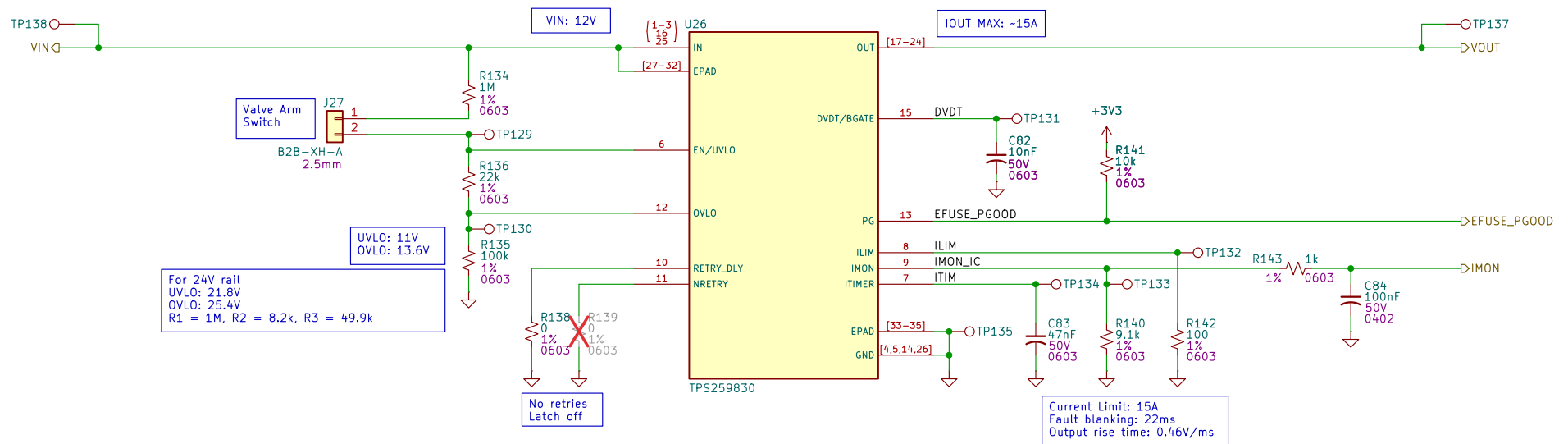
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EFUSE



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File: master_efuse.kicad_sch

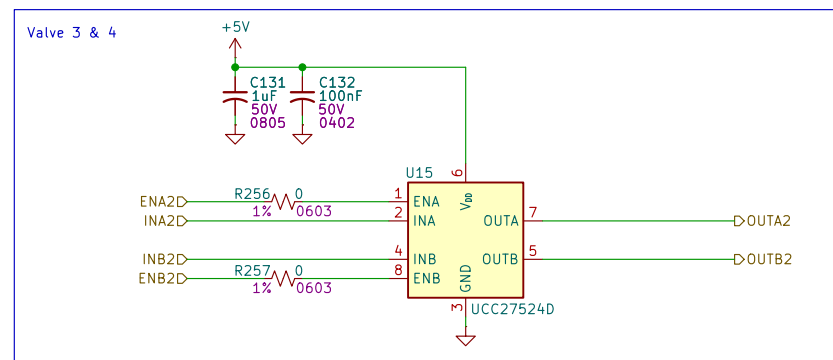
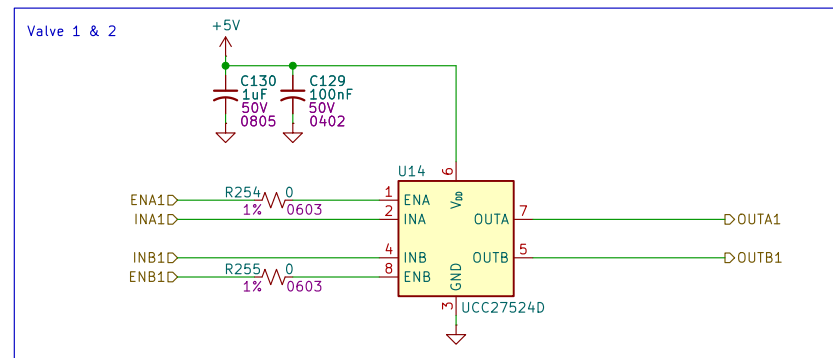
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Size: A4	Date: 2026-10-08
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SIZE: A1	
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Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4	Date: 2026-10-08
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Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Title: Valve Output

Title: Valve Output

Size: A4	Date:
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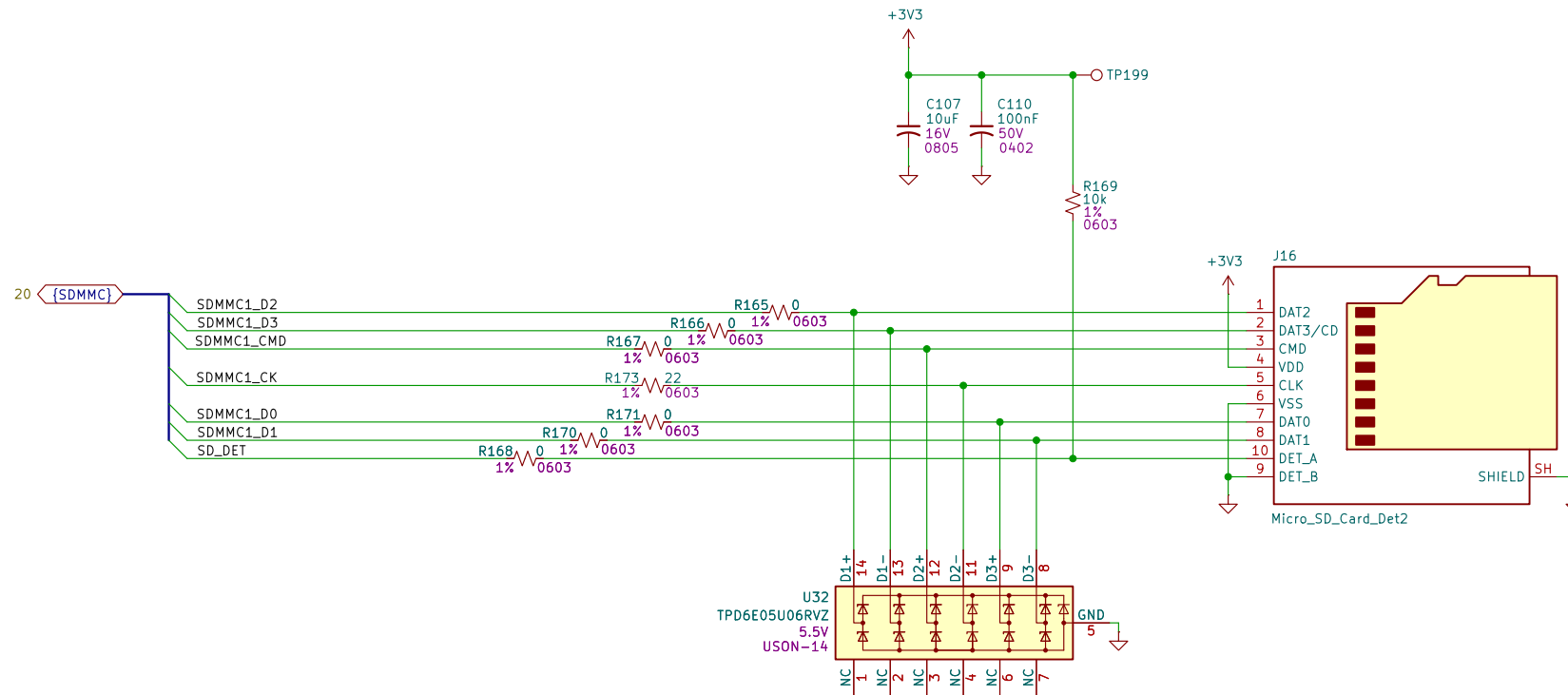
Date: 2026-10-08

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SD CARD



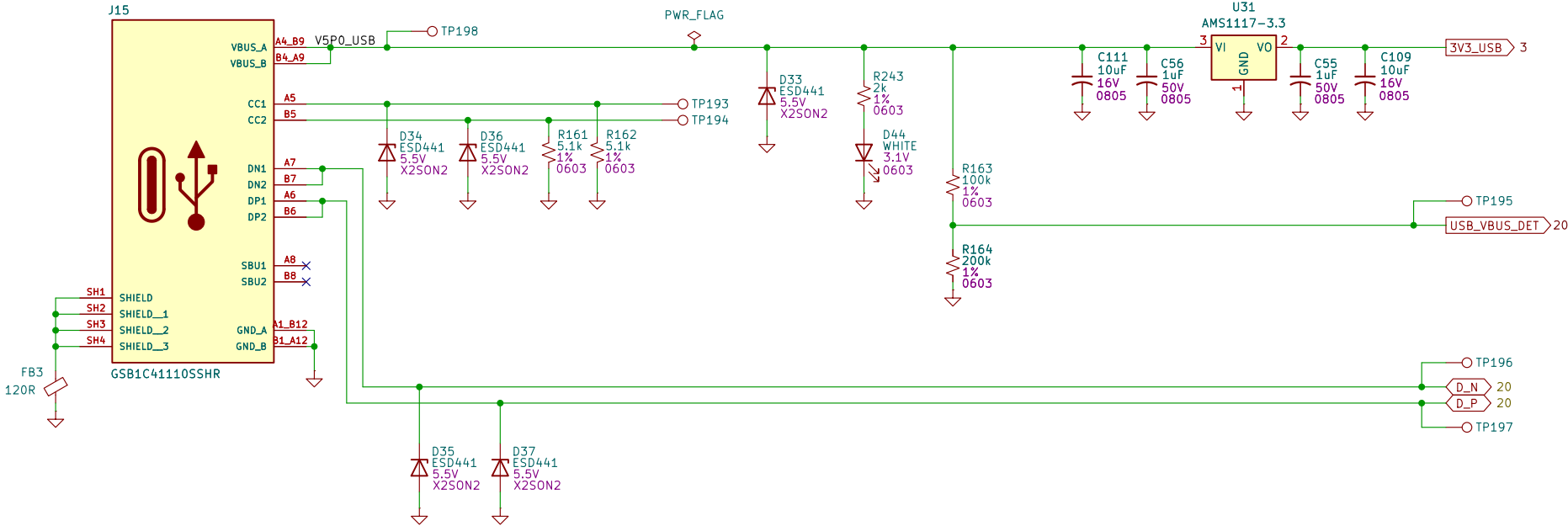
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File: sd_card.kicad_sch

Title: SD CARD

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

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USB-C



Sheet: /USB-C/
File: usb-c.kicad_sch

Title:

Size: A4

Date: 2026-10-08

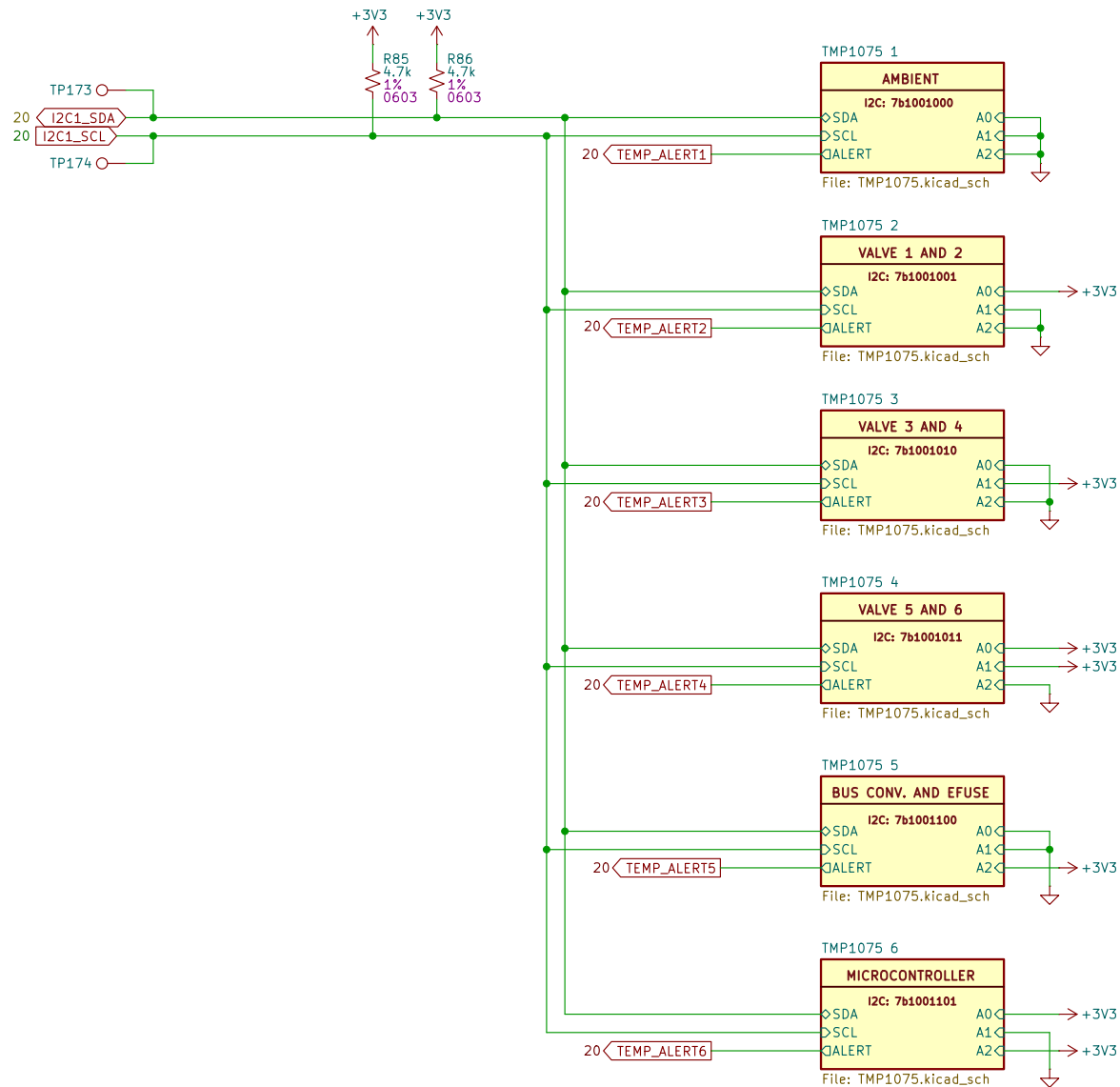
Rev: A

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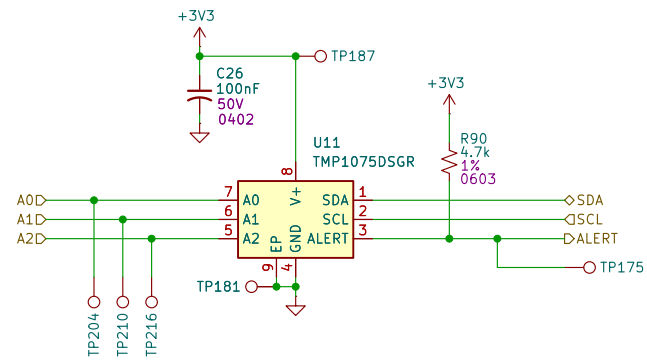
[illegible]

TEMPERATURE SENSORS



 STUDENT COMPETITION ROCKETRY TEAM		
Sheet: /Sensors/ File: sensors.kicad_sch		
Title: SENSORS		
Size: A4	Date: 2026-10-08	Rev: A
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TMP1075



Sheet: /Sensors/TMP1075 1/
File: TMP1075.kicad_sch

Title: TMP1075

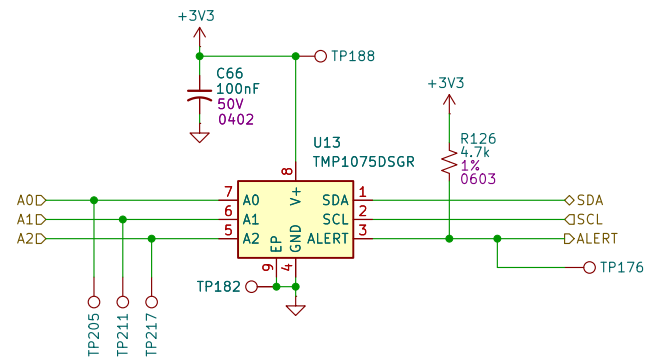
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TMP1075



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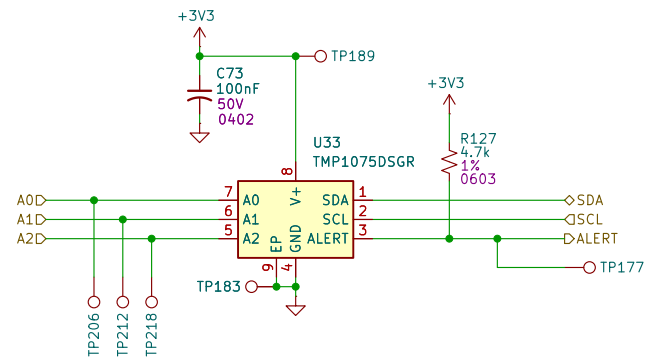
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Size: A4 Date: 2026-10-08

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TMP1075



Sheet: /Sensors/TMP1075 3/
File: TMP1075.kicad_sch

Title: TMP1075

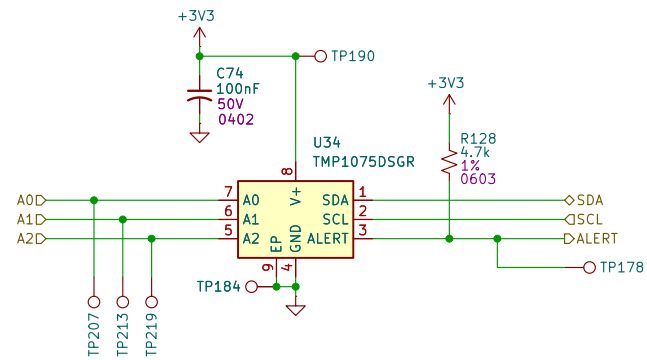
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TMP1075



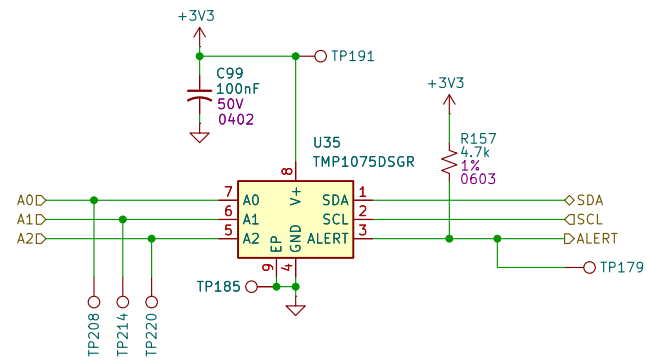
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Size: A4 Date: 2026-10-08
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TMP1075



Sheet: /Sensors/TMP1075 5/
File: TMP1075.kicad_sch

Title: TMP1075

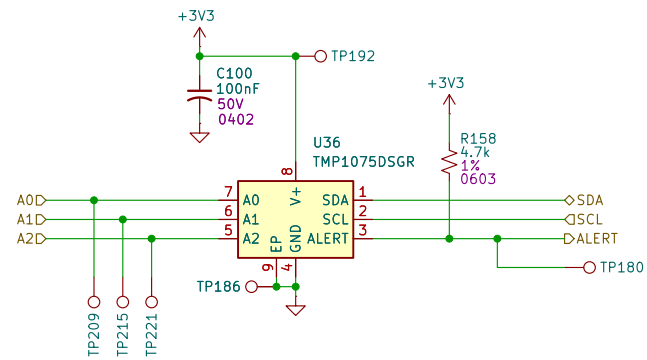
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KiCad E.D.A. 10.0.5

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TMP1075



Sheet: /Sensors/TMP1075 6/
File: TMP1075.kicad_sch

Title: TMP1075

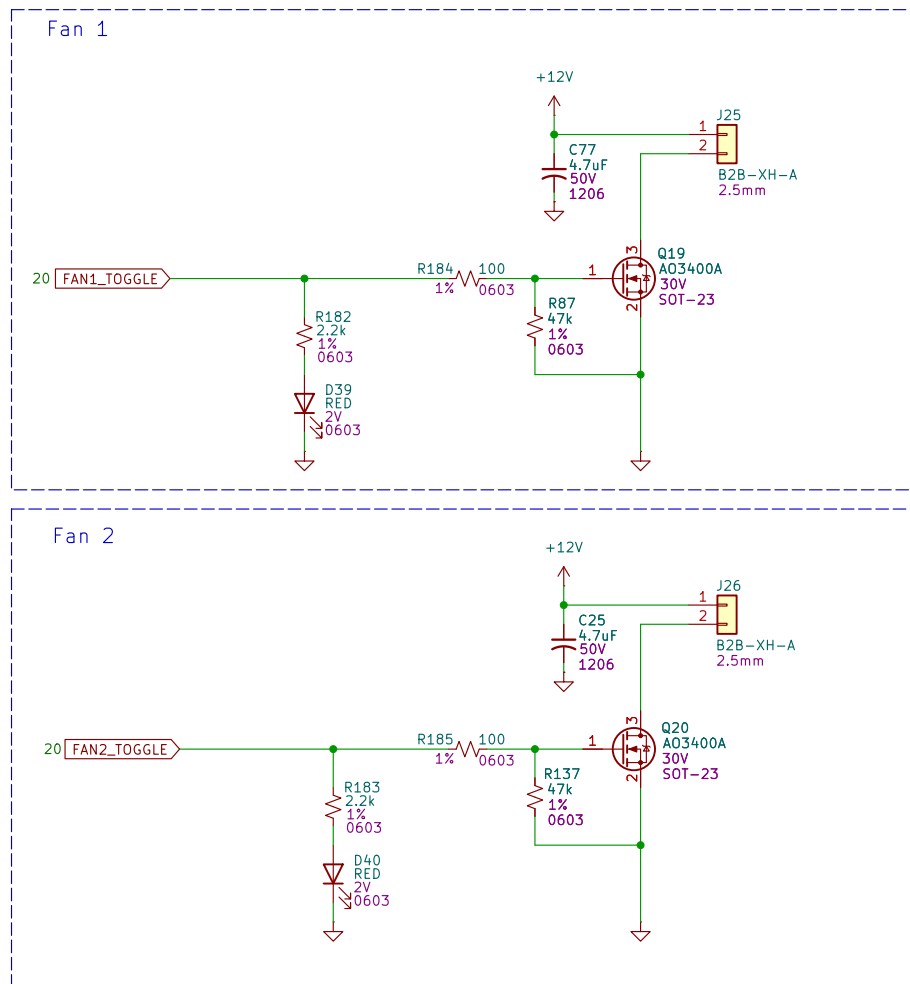
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KiCad E.D.A. 10.0.5

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FANS



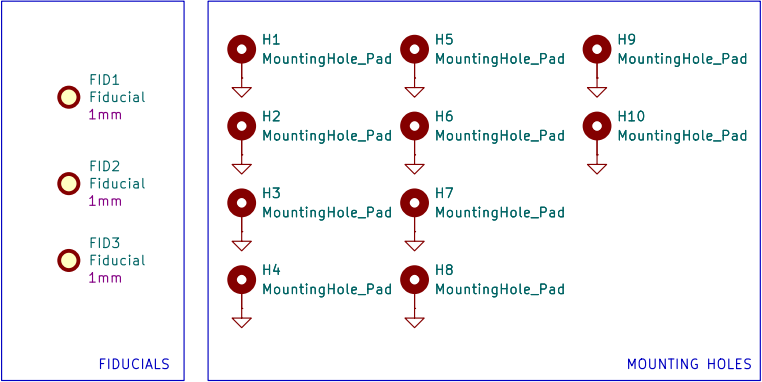
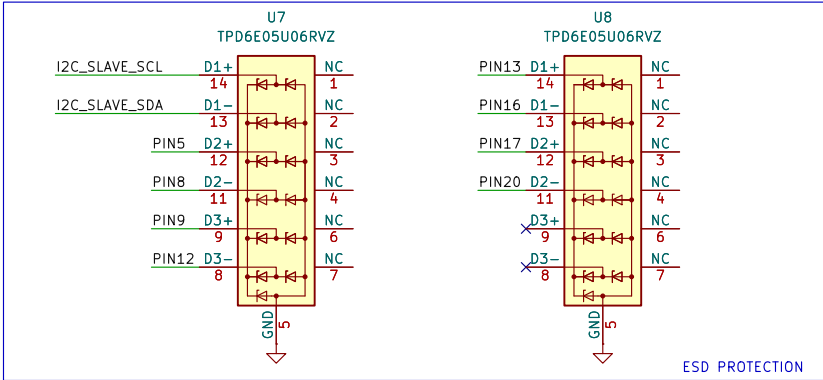
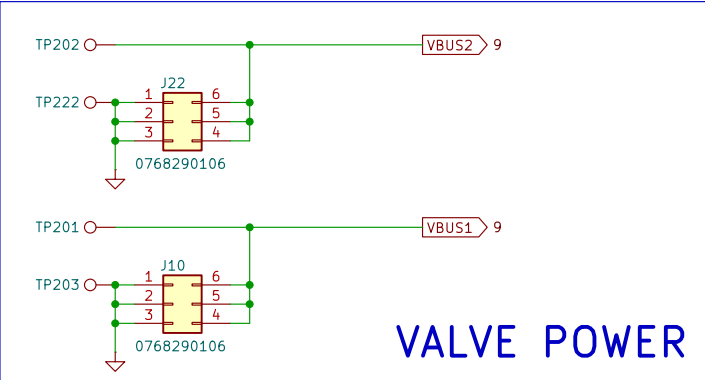
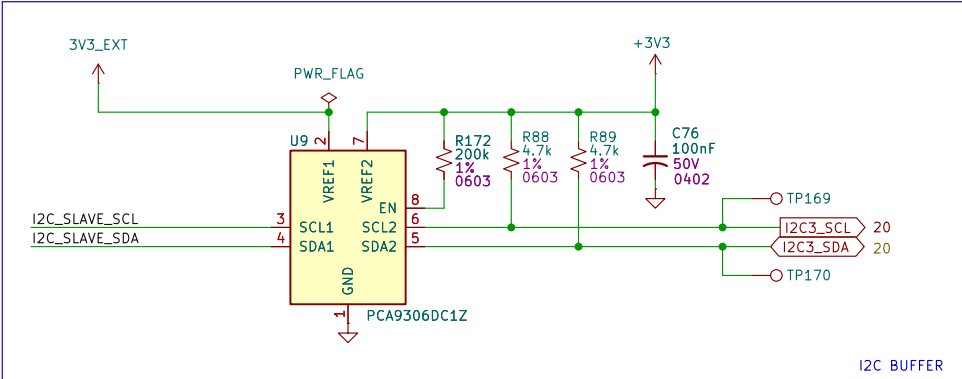
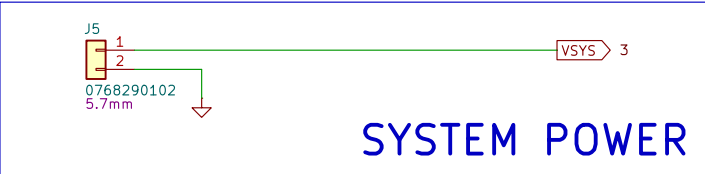
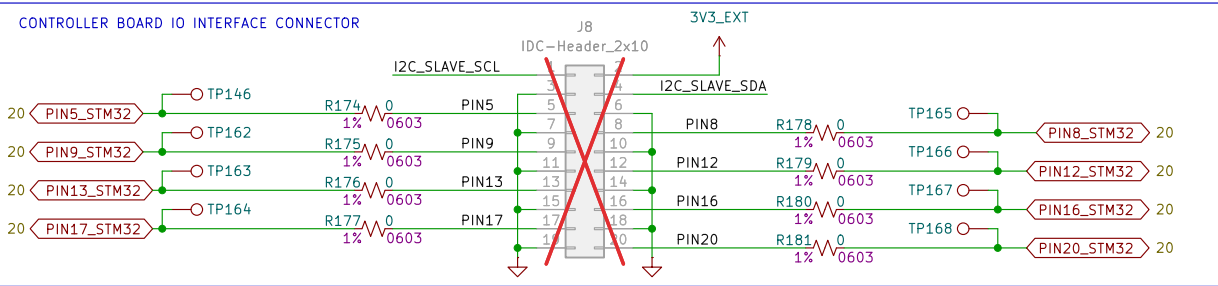
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Title: Fans

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

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MECHANICAL



Sheet: /Connectors/
File: connectors.kicad_sch

Title: MECHANICAL

Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-10-08

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